CASE STUDY REPORT

TRAFFIC ACCIDENT DATA ANALYSIS

BIG DATA ANALYTICS

Submitted in partial fulfilment of the requirements for the degree of

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1. EXECUTIVE SUMMARY

1.1 Overview and Motivation

Road traffic accidents are influenced by several factors, including road and weather conditions, lighting, vehicle type, time of occurrence, traffic volume and driver-related circumstances. Accident information may be stored across police records, transport department reports, emergency services and public datasets. Reviewing these records manually can make it difficult to identify recurring patterns and locations that may require road-safety attention.

Traffic Accident Data Analysis is a proposed Big Data Analytics project that organises accident records, analyses temporal and geographical patterns and examines factors associated with accident occurrence and severity. The workflow includes data ingestion, validation, preprocessing, exploratory analysis, feature engineering, analytical modelling, evaluation and dashboard-based visualisation. The system is intended to support evidence-based road-safety planning; it does not replace traffic authorities or establish the cause of an individual accident.

1.2 Objectives

• Collect or load an authorised traffic accident dataset with documented fields and usage terms.

• Validate accident dates, locations, severity labels, weather details and road-condition values.

• Clean missing, duplicated, inconsistent and invalid records using documented rules.

• Analyse accident patterns by time, location, road type, weather and severity.

• Develop baseline and suitable analytical or machine-learning methods for severity or hotspot analysis.

• Display accident trends, high-frequency locations and contributing indicators in a dashboard.

• Evaluate the system using suitable metrics and disclose limitations, bias risks and uncertainty.

1.3 Proposed Outcome

The expected prototype consists of a repeatable data-processing pipeline, descriptive analytics, an optional prediction or hotspot-analysis module and an interactive dashboard. It may show accident counts by month and hour, severity distribution, location-wise patterns and risk indicators. Actual model scores, hotspot results and prediction figures must be added only after implementation and testing on a specified dataset; this report does not invent experimental results.

2. PROBLEM STATEMENT AND ROAD-SAFETY CONTEXT

2.1 Problem Statement

Traffic accident data may be distributed across incident reports, road-safety databases and transport records. Manual review can be time-consuming and may not reveal repeated accidents at the same locations, peak accident periods or combinations of environmental and road-related indicators. The problem is to develop a structured analytics workflow that combines relevant accident records, identifies patterns and presents useful evidence for human review.

An observed association does not automatically prove causation. For example, a higher number of accidents during rainy periods may also be influenced by traffic volume, road usage and reporting practices. The system should therefore describe relationships in the available data and avoid claiming that one factor alone caused an accident.

2.2 Aim and Scope

The aim is to design and evaluate an offline prototype using historical traffic accident data. Candidate fields include accident date and time, approximate location, road type, weather, lighting, surface condition, vehicle category, collision type and recorded severity, depending on dataset availability. The project covers data validation, exploratory analytics, feature engineering, optional prediction, evaluation and visualisation.

The initial scope focuses on historical analysis and decision support. It does not provide emergency response, live vehicle tracking or automatic legal conclusions. Only public, authorised or appropriately anonymised data should be used. Personal identifiers, exact victim details and sensitive information should not be exposed in screenshots, reports or demonstrations.

2.3 Functional and Non-Functional Requirements

Functional requirements include importing supported files, validating columns and values, calculating accident summaries, analysing severity or hotspots, displaying explanations and exporting a summary. Non-functional requirements include privacy, reproducibility, readable visualisations, clear error messages, appropriate access control and transparent limitations. Analytical outputs should be reviewed by authorised road-safety personnel.

3. TRAFFIC ACCIDENT DATA AND SEVERITY INDICATORS

3.1 Candidate Data Fields

Potential input fields include accident identifier, date, time, latitude and longitude or an area code, road type, junction detail, speed limit, weather condition, road-surface condition, light condition, vehicle type, collision type, number of vehicles and recorded injury severity. The exact fields must be chosen based on what is available and permitted in the selected dataset.

Each field should have a documented definition, unit, valid range and source. Direct personal identifiers should be removed or masked. Location precision should be reduced where necessary to protect privacy, while retaining enough geographic detail for meaningful area-level analysis.

3.2 Defining Accident Severity

The target variable must be defined before model training. Depending on the source, the target may be a recorded severity category, such as fatal, serious injury or slight injury, or a numeric count such as the number of reported incidents in an area and period. The original dataset's definitions should be retained rather than creating unsupported categories.

Severity classification is not the same as predicting the exact outcome of a future crash. The model learns patterns represented in historical records, and important contextual factors may be missing. Outputs should be presented as estimates for planning and review.

3.3 Data Quality and Coverage

Traffic accident datasets can contain missing coordinates, duplicated incidents, inconsistent road names, invalid dates, unknown weather values and different severity coding systems. Missing information must not be silently treated as a real value. Records from one city, road network or period may not represent other regions. The report should document the source, date range, geographic coverage, number of records and known sampling limitations.

4. SYSTEM ARCHITECTURE

4.1 Architecture Overview

The proposed system contains six logical layers: data input, validation and cleaning, analytics processing, feature engineering, analysis or prediction, and dashboard/reporting. The input layer accepts public or authorised accident data. The validation layer checks schemas, value ranges, duplicates, missing fields and geographic validity. Python and Pandas can support small datasets, while Apache Spark/PySpark may be used when record volume justifies distributed processing.

The analysis layer calculates trends, severity distributions and location-based summaries. An optional prediction layer applies a documented baseline and selected model. The dashboard presents results for review. Original data should remain unchanged. Cleaned data, derived features, model files and evaluation outputs should be stored separately so that each analysis run can be reproduced.

4.2 Architecture Diagram

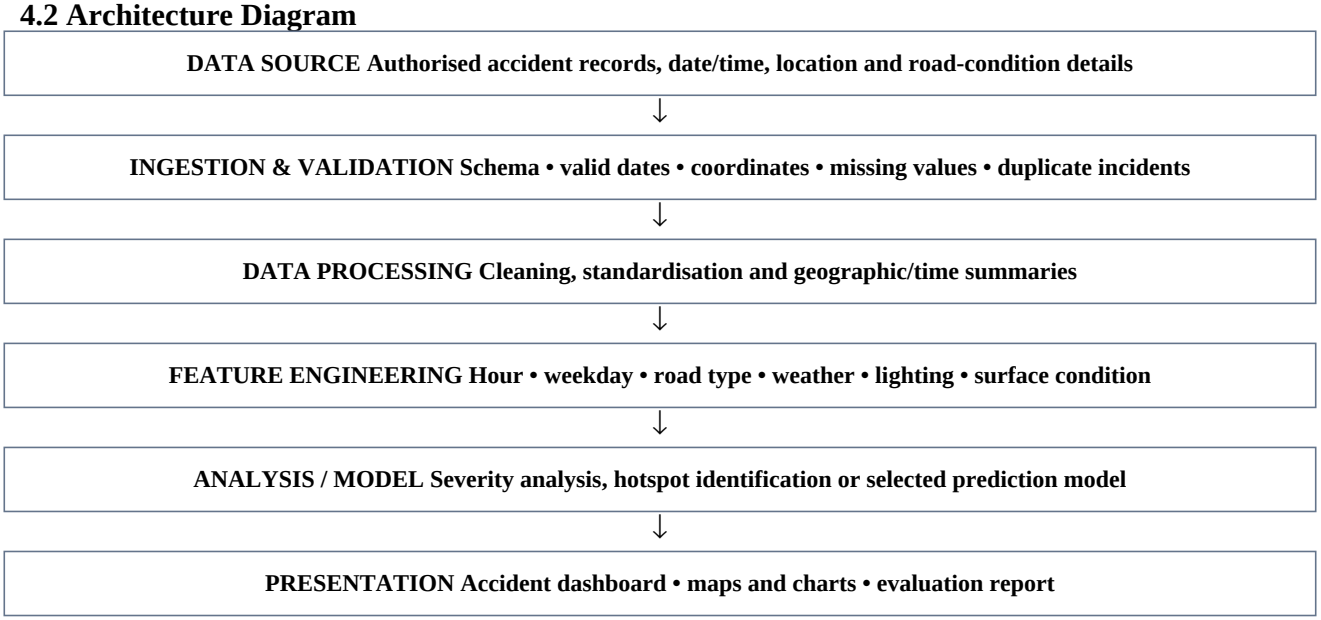

FIGURE 1. TRAFFIC ACCIDENT ANALYSIS SYSTEM ARCHITECTURE

4.3 Module Description

Data ingestion module: reads authorised CSV or spreadsheet data and records the schema. Validation module: checks identifiers, dates, coordinates and categorical values. Analytics module: calculates accident counts, severity distributions and time trends. Feature module: derives time bands, location groupings and combined road-condition indicators. Analysis module: identifies patterns or estimates a defined target. Presentation module: displays aggregated trends, maps and evaluation results.

4.4 End-to-End Data Flow

The user selects an authorised dataset and the system validates its structure. Invalid or ambiguous values are flagged for correction or documented treatment. Clean records are transformed into analysis-ready features. Exploratory analysis describes accident patterns, and any model is trained using a development subset. Evaluation uses held-out records that were not used to fit the model. The dashboard displays results with explanations and data limitations.

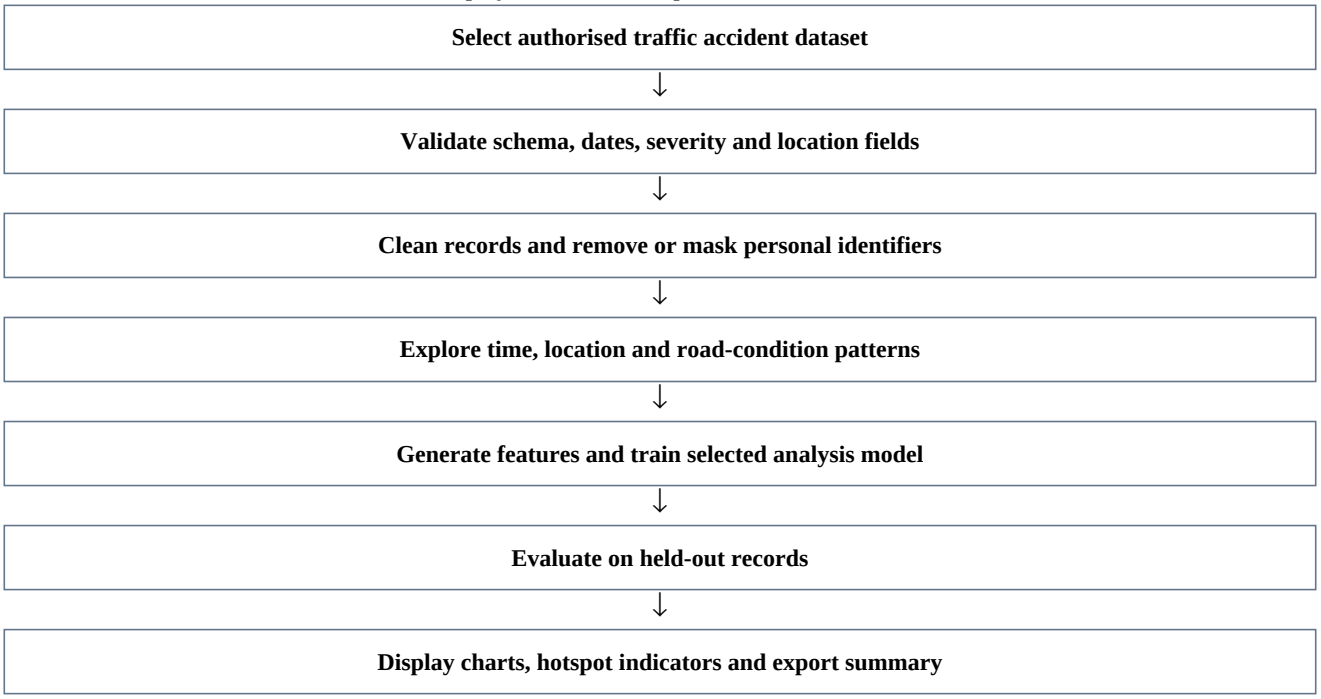

FIGURE 2. END-TO-END DATA FLOW

5. DATASET INGESTION AND PREPROCESSING

5.1 Dataset Selection

The project should use an authorised public traffic accident dataset or an institutional dataset with documented terms of use. The report should identify the source, version, date range, geographic coverage, number of records, severity definition and permitted fields. If a synthetic dataset is used for demonstration, it must be labelled as synthetic and must not be presented as real accident evidence.

5.2 Preprocessing Procedure

• Read the source and record its schema, row count, date range and geographic coverage.

• Standardise column names, date formats, time formats and location labels.

• Validate coordinates, dates, severity values and numeric ranges.

• Identify missing values, duplicate incidents and inconsistent category labels.

• Apply documented imputation, exclusion or review rules; do not silently invent missing values.

• Remove or pseudonymise personal identifiers and minimise location precision where needed.

• Split data before fitting transformations or models to avoid data leakage.

5.3 Data Quality Checks

Checks should detect impossible dates, coordinates outside the expected region, duplicate incident records, invalid severity labels and inconsistent road-condition categories. Unexpected values should be flagged rather than silently corrected. A data-quality summary should report the number of records read, retained, corrected and excluded.

6. EXPLORATORY TRAFFIC ACCIDENT ANALYTICS

6.1 Descriptive Analysis

Exploratory analysis describes accident patterns before prediction. Useful summaries include accident counts by month, weekday and hour; severity distribution; location-wise incident frequency; road-type comparison; weather and lighting distributions; and trends across years. The analysis should also inspect missingness, geographic coverage and the number of records available for each category.

Group-level patterns can help identify locations or periods that may deserve further road-safety review. However, raw accident counts do not account for traffic exposure. A road with more vehicles may have more reported accidents simply because more people use it. Where reliable traffic-volume data are available, exposure-adjusted rates may provide additional context.

6.2 Recommended Charts

• Bar chart: accident count by road type or severity category.

• Histogram: distribution of accidents by hour of day.

• Line chart: monthly or yearly accident trends.

• Heatmap: accident frequency by weekday and hour.

• Map: aggregated accident locations or hotspot areas, where coordinates are reliable.

• Stacked bar chart: severity distribution by weather, lighting or road condition.

• Summary cards: total records, date range, severe-incident count and records with missing locations.

Charts should use readable axes and state the period and geographic area. Maps should aggregate or blur locations when exact coordinates could identify people or sensitive sites. Counts and rates must be labelled clearly.

7. FEATURE ENGINEERING

7.1 Candidate Features

Feature engineering converts raw accident records into variables suitable for analysis. Candidate features include hour band, weekday, month, weekend indicator, road category, junction indicator, weather group, lighting condition, surface condition, vehicle category and geographic area. Features must be derived only from fields available in the selected dataset and documented with valid values.

7.2 Preventing Data Leakage

Data leakage occurs when a model uses information that would not have been available at prediction time. If the goal is to estimate severity using conditions known at the scene, post-incident investigation outcomes should not be included as input features. Preprocessing transformations must be fitted on training data only. For time-based analysis, earlier periods should generally be used for training and later periods for testing.

8. ACCIDENT SEVERITY AND RISK ANALYSIS METHODS

8.1 Baseline Method

A baseline provides a simple reference for evaluating a more complex model. For severity classification, a majority-class baseline or a documented rule-based approach can provide a point of comparison. For accident-count estimation, a simple historical average may be used. The baseline and advanced model must be evaluated on the same held-out records.

8.2 Machine-Learning Models

Potential models include Logistic Regression, Decision Tree or Random Forest Classifier for severity classification, and Linear Regression, Random Forest Regressor or suitable count models for accident-count estimation. The selected task must be explicit: classification predicts a category, while regression predicts a numeric value. Model selection should consider dataset size, interpretability, class imbalance and the ability to explain results to road-safety reviewers.

The initial prototype should focus on one baseline and one suitable model. Hyperparameters should be chosen using training and validation data, not the final test set. Model outputs should be presented as estimates rather than guaranteed outcomes.

8.3 Analysis Flowchart

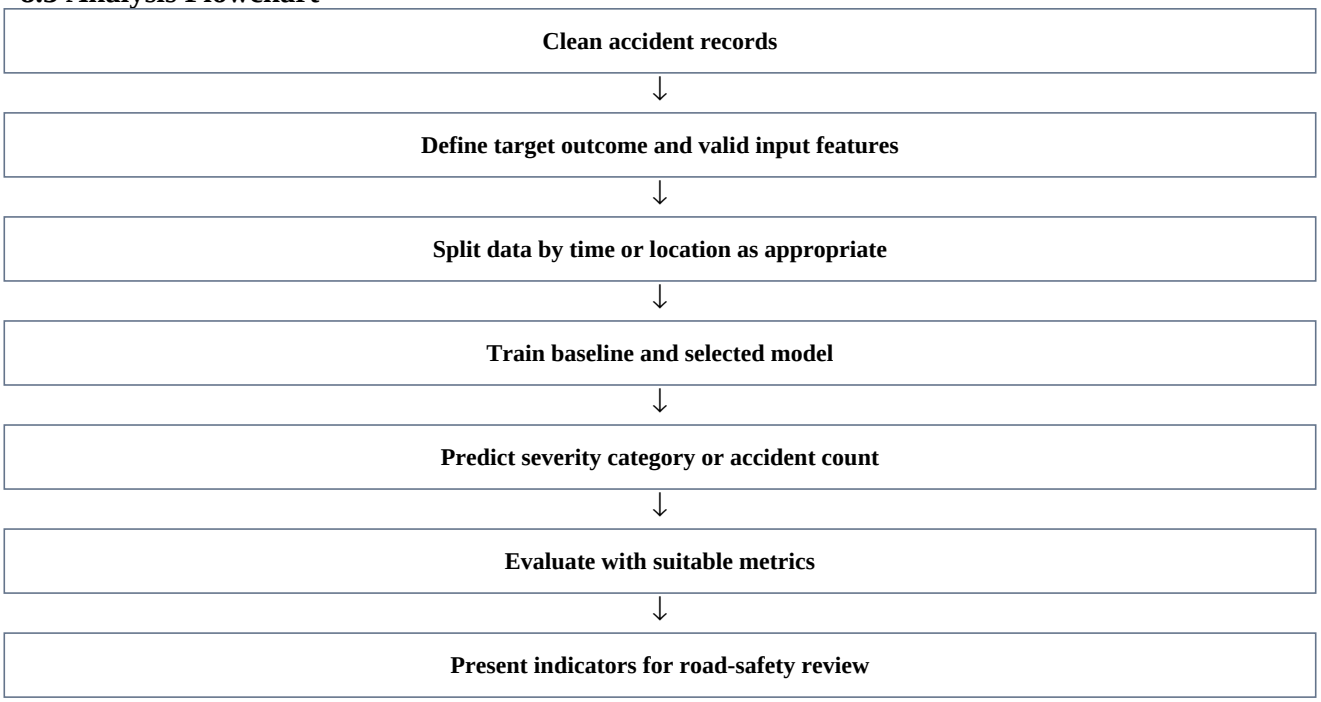

FIGURE 3. ACCIDENT ANALYSIS FLOWCHART

9. RISK INTERPRETATION AND ROAD-SAFETY ACTION

9.1 Risk Indicator Design

If areas or periods are grouped into risk bands, the thresholds should be defined before evaluation and grounded in documented criteria. A high historical count should not automatically be interpreted as a high individual risk. Indicators should communicate the period, geographic unit, data coverage and uncertainty. Risk patterns may change as traffic conditions, road design and reporting practices change.

9.2 Human Review and Support

A hotspot indicator may prompt a review of signage, lighting, road surface, junction design, speed management or traffic flow. Traffic authorities should verify the source records and consider traffic exposure and local context before taking action. The system should not automatically assign blame, issue penalties or make legal conclusions.

10. DASHBOARD, REPORTING AND USER WORKFLOW

10.1 Dashboard Layout

The dashboard may contain summary cards, accident trends, severity charts, time-based heatmaps, road-condition comparisons and an aggregated location map. Filters can include date range, district or area, road type, severity and weather condition. Each chart should display the active filters and the number of records included. The interface should clearly distinguish missing data from zero counts.

10.2 User Workflow

• Select an authorised traffic accident dataset.

• Validate incident, date, location, severity and condition fields.

• Clean records and display data-quality warnings.

• Explore accident trends by time, place and road condition.

• Define the target outcome and construct valid features.

• Train a baseline and a selected model, if prediction is included.

• Evaluate predictions on held-out data.

• Display trends and risk indicators with explanations and limitations.

• Export an aggregated summary for authorised review.

10.3 Reporting and Access Control

An exported report should include the data source, date range, geographic coverage, preprocessing rules, target definition, train/test split strategy, model settings and evaluation metrics. Personal identifiers should be removed, and precise coordinates should be restricted where appropriate. Demonstration screenshots should use aggregated or anonymised information.

11. BIG DATA PROCESSING WITH APACHE SPARK

11.1 Why Spark May Be Relevant

Traffic accident analytics can involve large collections of incidents across multiple years, districts and road networks. Apache Spark supports distributed transformations and aggregations across partitions. PySpark can be used to standardise records, combine data sources, calculate time and location summaries and generate analytical features. For a small classroom dataset, Pandas may be sufficient; Spark should be included when data size or learning objectives justify it.

11.2 Runtime and Scalability Evaluation

Performance claims should be based on measured experiments. Record hardware, dataset size, row count, partition settings, software versions and runtime. Compare equivalent operations and specify whether file-loading time is included. Distributed processing does not automatically improve analytical accuracy; it primarily changes how data transformations are performed at scale.

12. DATASET PREPARATION AND TEST DESIGN

12.1 Test Strategy

The evaluation design should match the intended task. If the goal is to analyse future accident patterns, training should use earlier periods and testing should use later periods. If the goal is to generalise to new geographic areas, location-level separation should be considered. Class imbalance must be inspected when severity categories are used. All preprocessing and model tuning should be performed without using final test labels.

12.2 Test Case Matrix

Test case	Expected check
Valid accident data	Schema and summary statistics are generated
Invalid date or coordinate	Invalid values are flagged
Missing severity field	Missing value is reported and handled by documented rule
Duplicate incident	Duplicate records are identified
Unknown road/weather category	Unrecognised category is flagged
Leakage check	Unavailable post-incident information is excluded
Imbalanced severity classes	Class distribution and suitable metrics are reported
Empty or malformed input	Clear validation error is shown
Larger dataset	Runtime and resource conditions are recorded

Table 1. Test Case Matrix

12.3 Reproducibility

Record dataset source and version, date range, geographic coverage, target definition, feature list, split strategy, model parameters, random seed where applicable and software environment. Preserve a summary of rows corrected or excluded. These details allow another student or evaluator to reproduce the analysis and understand differences in the results.

13. EVALUATION METRICS AND ERROR ANALYSIS

13.1 Metrics for Accident Count Prediction

For regression, Mean Absolute Error (MAE) describes the average absolute difference between predicted and observed counts. Root Mean Squared Error (RMSE) gives greater weight to larger errors. R-squared may be included as a complementary measure but should not replace error analysis. Report the count scale and number of test records alongside the metrics.

13.2 Metrics for Severity Classification

For classification, precision measures the proportion of records predicted as a class that belong to that class according to reference labels, while recall measures the proportion of actual class records detected. F1-score balances precision and recall. A confusion matrix helps show false positives and false negatives. Accuracy alone can be misleading when severity classes are imbalanced.

No measured performance figures are claimed in this report because the prototype has not been executed against a specified dataset. Actual metrics should be added only after repeatable testing. The report should describe whether records are real, public or synthetic and state the limits of generalisation.

14. LIMITATIONS, RISKS AND MITIGATIONS

14.1 Limitations

Analytical quality depends on dataset size, reporting coverage, location accuracy, severity definitions and changes in road infrastructure or traffic volume. A model trained in one city or period may not generalise to another. Reported accident counts may reflect differences in reporting practices. Without exposure information such as traffic volume or distance travelled, raw counts cannot fully describe risk.

Privacy risks arise because incident records may contain personal or precise location information. Missing data, imbalanced severity labels and inaccurate coordinates can produce misleading outputs. Historical association does not prove causation, and synthetic data cannot demonstrate real-world predictive performance.

14.2 Mitigations

Use authorised and minimised data; remove or mask personal identifiers; document source and target definitions; validate dates and coordinates; prevent leakage; evaluate with appropriate metrics; review false positives and false negatives; restrict dashboard access; and present outputs as decision-support indicators. Do not use the prototype to assign blame or automate legal decisions.

15. EXAMPLE USE CASE: IDENTIFYING HIGH-RISK ROAD CONDITIONS

15.1 Illustrative Scenario

An authorised road-safety analyst loads an anonymised dataset containing accident dates, approximate locations, severity and road-condition details for a completed period. The system validates the fields, identifies missing coordinates and duplicate incidents, and then displays area-wise counts and time patterns. The analyst reviews locations with repeated reported incidents and checks the relevant road, weather and lighting indicators.

The dashboard presents the data coverage and limitations. The analyst verifies the source records and considers traffic volume, road design and local conditions before recommending a site inspection. This scenario illustrates the workflow and is not a claim that real locations were classified.

15.2 Expected Prototype Output

The expected demonstration includes an accident summary, monthly trend chart, severity distribution, hour-wise heatmap, road-condition comparison, aggregated hotspot map and data-quality summary. If a prediction model is implemented, model metrics and an anonymised evaluation summary should also be shown. Actual screenshots, data counts and measured scores should be inserted after the prototype has been implemented and tested.

16. FUTURE ENHANCEMENTS

Future versions may include reliable traffic-volume integration, near-real-time authorised data feeds, road-network mapping, explainability methods, careful model recalibration and comparison across districts and years. Additional work could evaluate whether hotspot indicators remain stable over time and whether road-safety interventions are followed by measurable changes. Any deployment should include privacy controls, access logging, clear communication about model use and human review.

17. CONCLUSION

Traffic Accident Data Analysis proposes a structured Big Data Analytics workflow for understanding historical road accident patterns and identifying locations, periods and conditions that may deserve further safety review. The design combines data validation, exploratory analytics, feature engineering, optional prediction, evaluation and dashboard-based visualisation. Apache Spark/PySpark can support distributed processing when dataset size justifies it.

The usefulness of the project depends on authorised, high-quality data, reliable location and severity definitions and a leakage-free evaluation. The final demonstration should document the dataset, show data-quality handling, compare a baseline with the selected method and report measured results. Outputs must be interpreted cautiously and reviewed by road-safety

personnel; they should support evidence-based planning rather than serve as automatic conclusions.

APPENDIX A – SUGGESTED PROTOTYPE WORKFLOW

• Input layer: reads authorised accident data.

• Preprocessing layer: validates dates, coordinates, duplicates and missing values.

• Analytics layer: calculates time, severity and location summaries.

• Feature layer: creates time, road and environmental indicators.

• Analysis layer: applies descriptive analysis and an optional baseline/model.

• Dashboard layer: presents trends, maps and anonymised summaries.

• Evaluation layer: records test design, metrics and limitations.

A.2 Implementation Sequence

• Select an authorised dataset and define the analysis target.

• Inspect the schema, date range, severity codes and geographic coverage.

• Build validation and preprocessing rules.

• Create exploratory charts and a data-quality report.

• Construct features available at analysis time.

• Train a baseline and one suitable model if prediction is required.

• Evaluate on held-out data using suitable metrics.

• Build the dashboard and document privacy safeguards and limitations.

APPENDIX B – EVALUATION CHECKLIST AND GLOSSARY

• Record dataset source, version, date range and permitted use.

• Define the target outcome before evaluation.

• Preserve original data and document cleaning rules.

• Validate coordinates, dates and severity labels.

• Prevent leakage and split records appropriately.

• Compare the selected model with a baseline.

• Report MAE/RMSE for regression or precision/recall/F1 for classification.

• Review false positives, false negatives and class imbalance.

• Mask personal identifiers and sensitive coordinates in screenshots.

• Document limitations and require human review of risk indicators.

Glossary: Accident severity: recorded level of harm associated with an incident. Feature engineering: deriving analytical variables from raw accident records. Data leakage: use of information unavailable at prediction time. Hotspot: an area with a relatively high concentration of recorded incidents under a stated method. Regression: prediction of a numeric value. Classification: prediction of a category. Precision: proportion of predicted positive cases that are correct. Recall: proportion of actual positive cases detected. PySpark: Python interface for Apache Spark distributed data processing.